

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement records the complete prospective-service registry, numerical semantics, confidence intervals, datewise error accounts and source-preservation map for Neural Bellman Operators. The active mathematical statements are proved in the article; all preceding proofs and applications remain in the complete companions.

1. EXECUTION, NUMERICAL SEMANTICS AND COMPLETE RECORDS

The active article retains the constructive own-future NBO operator and adds trained neural action search, non-tensor verification, prospective economic stopping and validation-qualified learned structure. Its mathematical statements are proved in the article. This supplement records the execution contract and all service outcomes. The complete development and proof companions retain every preceding theorem, application and adverse experiment under its original assumptions.

1.1. *Source and outcome identity*

The controlling advisory report reviews manuscript commit 00e837adb431f4d4b5248fc6d5ff1fc927dd3b65. Its review commit is adf1256cff9cde365246a3db2dac90c72fda3b13. The separately frozen primary and signed-sensitivity executions are inherited from 7f4de488134e93b6cfb5dd95a3e6d7e14b45ac2a. This manuscript integration does not rerun those production services or add cost samples. The source freezes, interrupted controller, corrected controller, fitted parameters, raw endpoints, clocks and immutable process checkpoints remain unchanged.

The publication audit reloads each saved fitted critic, reconstructs its proposal, and regenerates each unique whole-domain certificate with the frozen numerical algorithms. Every saved candidate endpoint, covering endpoint, residual band, value band and policy choice is compared with that calculation. Repetition-equivalent inputs are computed once and reused only after their canonical input identities match. This is primitive re-evaluation of the recorded certificate with the same

algorithm, not an independent proof assistant or an independent implementation of the entire economic simulator.

All 132 services are then checked for their exact incumbent transfer, nine common candidates and tenth proposal, complete continuous-action cover, strict minimum-upper selection, feasibility, exact finite-sweep recurrence, declared look order, cumulative sample prefixes and first legal stopping decision. Every referenced file and process-log hash is checked. No later inference or construction is permitted after a target crossing. A finite budget exhaustion remains in the outcome table.

1.2. *Cross-platform moment verification*

Different numerical-library versions can differ in the final rounding of a sample second moment even when their endpoint arrays are byte identical. The audit therefore does not overwrite a scientific interval with a newly rounded interval and call it the same record. Instead, it converts each binary64 endpoint to an exact rational, computes its exact sample mean, second moment and unbiased variance, and checks their containment in the originally stored outward moment account. It reconstructs the rational empirical-Bernstein endpoints from those stored accounts. Thus a valid original interval can be verified without assuming bit identity of a library reduction on another platform.

Validation confidence boxes receive the corresponding exact-mean check, with their stored confidence radius and numerical-bin allowance. All policy-cost endpoints retain signed path gains. Conditional expected-cost safety does not imply that a common-shock pathwise gain is nonnegative; such paths are not clipped away. The declared independent-bin statistical model remains distinct from the reproducibility of a pseudorandom seed.

1.3. *Residual cache and signed tubes*

A range query includes every closed leaf meeting the query box. This is conservative at a boundary but preserves the half-open deployed observation rule. No smoothness is imposed on the actor. The first-stage installed policy is globally zero. Sensitivity is used only under that verified reference identity; a changed incumbent

falls back to the generic cache. The forward zero-policy tube contains the entire future innovation support and the backward tube uses the sparse Jacobian of the original drift. Its first innovation is integrated over the declared bins with exact probabilities. The algorithm does not recursively enumerate all q^T future bins.

Each raw candidate and covering comparison retains its base-cache interval, signed-tube interval and intersection. Empty intersections are errors, not opportunities to choose a midpoint. Table entries for $U - C$ and $C - L$ are maxima of independently nonnegative cellwise quantities. Their separate maxima need not sum to the displayed maximum of $U - L$. Refinement attribution is a comparison of valid enclosures, not a causal decomposition of state discretization, action search, stochastic integration and floating-point error.

1.4. *Exact neural action search*

The new scalar-action solver reads the exact stored dyadic coefficients. It enumerates rational hinge breakpoints, constructs the cubic derivative of each quartic piece, isolates every real stationary root by rational intervals, and compares candidate lattice actions by exact rational objective values. Multiple roots and zero innovation projections have explicit tests. The retained saved-model regressions specify one acquired-cell center in each task before checking its entire finite lattice exhaustively. They create neither a replacement production policy nor a new economic observation. Root-isolation cost and coefficient bit lengths are not replaced by a unit-cost count in a runtime claim.

2. COMPLETE PROSPECTIVE-SERVICE REGISTRY

TABLE I: Every prospective service and timing repetition

Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)
P 2 2	W	49/50	0	1	16384	H	1.149		
P 2 2	W	49/50	1	1	16384	H	1.144		
P 2 2	W	49/50	2	1	16384	H	1.153		
P 2 2	W	9/10	0	2	65536	H	4.664		

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3		P 2	2	W	9/10	1	2	65536	H	4.617	3
4		P 2	2	W	9/10	2	2	65536	H	4.621	4
5		P 2	2	E	49/50	0	1	16384	H	2.719	5
6		P 2	2	E	49/50	1	1	16384	H	2.714	6
7		P 2	2	E	49/50	2	1	16384	H	2.723	7
8		P 2	2	E	9/10	0	2	65536	H	10.442	8
9		P 2	2	E	9/10	1	2	65536	H	10.313	9
10		P 2	2	E	9/10	2	2	65536	H	10.247	10
11		P 4	4	E	49/50	0	2	16384	H	37.304	11
12		P 4	4	E	49/50	1	2	16384	H	36.875	12
13		P 4	4	E	49/50	2	2	16384	H	37.219	13
14		P 4	4	E	9/10	0	2	16384	E	35.003	14
15		P 4	4	E	9/10	1	2	16384	E	35.275	15
16		P 4	4	E	9/10	2	2	16384	E	35.054	16
17		P 8	6	E	49/50	0	2	65536	E	47.362	17
18		P 8	6	E	49/50	1	2	65536	E	48.146	18
19		P 8	6	E	49/50	2	2	65536	E	47.742	19
20		P 8	6	E	9/10	0	2	4096	E	43.171	20
21		P 8	6	E	9/10	1	2	4096	E	44.311	21
22		P 8	6	E	9/10	2	2	4096	E	43.425	22
23		P 2	2	Q	49/50	0	1	16384	H	0.936	23
24		P 2	2	Q	49/50	1	1	16384	H	0.933	24
25		P 2	2	Q	49/50	2	1	16384	H	0.928	25
26		P 2	2	Q	9/10	0	2	65536	H	3.900	26
27		P 2	2	Q	9/10	1	2	65536	H	3.904	27
28		P 2	2	Q	9/10	2	2	65536	H	3.879	28
29		P 4	4	Q	49/50	0	2	16384	H	10.739	29
30		P 4	4	Q	49/50	1	2	16384	H	10.567	30
31		P 4	4	Q	49/50	2	2	16384	H	10.715	31
32		P 4	4	Q	9/10	0	2	16384	E	9.153	32
		P 4	4	Q	9/10	1	2	16384	E	8.714	
		P 4	4	Q	9/10	2	2	16384	E	8.462	
		P 8	6	Q	49/50	0	2	65536	E	17.204	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3	P	8	6	Q	49/50	1	2	65536	E	16.978	3
4	P	8	6	Q	49/50	2	2	65536	E	17.123	4
5	P	8	6	Q	9/10	0	2	4096	E	13.132	5
6	P	8	6	Q	9/10	1	2	4096	E	13.187	6
7	P	8	6	Q	9/10	2	2	4096	E	13.249	7
8	P	2	2	N	49/50	0	1	16384	H	2.372	8
9	P	2	2	N	49/50	1	1	16384	H	2.376	9
10	P	2	2	N	49/50	2	1	16384	H	2.381	10
11	P	2	2	N	9/10	0	2	65536	H	7.159	11
12	P	2	2	N	9/10	1	2	65536	H	7.114	12
13	P	2	2	N	9/10	2	2	65536	H	7.112	13
14	P	4	4	N	49/50	0	2	16384	H	18.806	14
15	P	4	4	N	49/50	1	2	16384	H	18.506	15
16	P	4	4	N	49/50	2	2	16384	H	18.913	16
17	P	4	4	N	9/10	0	2	16384	E	17.163	17
18	P	4	4	N	9/10	1	2	16384	E	17.608	18
19	P	4	4	N	9/10	2	2	16384	E	17.196	19
20	P	8	6	N	49/50	0	2	65536	E	23.934	20
21	P	8	6	N	49/50	1	2	65536	E	24.386	21
22	P	8	6	N	49/50	2	2	65536	E	23.957	22
23	P	8	6	N	9/10	0	2	4096	E	19.988	23
24	P	8	6	N	9/10	1	2	4096	E	20.054	24
25	P	8	6	N	9/10	2	2	4096	E	19.783	25
26	P	2	2	F	49/50	0	1	16384	H	1.202	26
27	P	2	2	F	49/50	1	1	16384	H	1.195	27
28	P	2	2	F	49/50	2	1	16384	H	1.195	28
29	P	2	2	F	9/10	0	2	65536	H	4.761	29
30	P	2	2	F	9/10	1	2	65536	H	4.729	30
31	P	2	2	F	9/10	2	2	65536	H	4.708	31
32	S	2	2	W	49/50	0	1	4096	H	0.727	32
	S	2	2	W	49/50	1	1	4096	H	0.739	
	S	2	2	W	49/50	2	1	4096	H	0.888	
	S	2	2	W	9/10	0	1	16384	H	0.808	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)		
1											1	
2											2	
3		S	2	2	W	9/10	1	1	16384	H	0.781	3
4		S	2	2	W	9/10	2	1	16384	H	1.345	4
5		S	2	2	E	49/50	0	1	4096	H	1.620	5
6		S	2	2	E	49/50	1	1	4096	H	1.865	6
7		S	2	2	E	49/50	2	1	4096	H	1.688	7
8		S	2	2	E	9/10	0	1	16384	H	1.740	8
9		S	2	2	E	9/10	1	1	16384	H	1.678	9
10		S	2	2	E	9/10	2	1	16384	H	1.737	10
11		S	4	4	E	49/50	0	1	4096	H	11.605	11
12		S	4	4	E	49/50	1	1	4096	H	11.700	12
13		S	4	4	E	49/50	2	1	4096	H	11.589	13
14		S	4	4	E	9/10	0	1	4096	H	11.781	14
15		S	4	4	E	9/10	1	1	4096	H	11.515	15
16		S	4	4	E	9/10	2	1	4096	H	11.684	16
17		S	8	6	E	49/50	0	1	4096	H	31.428	17
18		S	8	6	E	49/50	1	1	4096	H	31.032	18
19		S	8	6	E	49/50	2	1	4096	H	31.304	19
20		S	8	6	E	9/10	0	1	16384	H	31.833	20
21		S	8	6	E	9/10	1	1	16384	H	31.886	21
22		S	8	6	E	9/10	2	1	16384	H	35.303	22
23		S	2	2	Q	49/50	0	1	4096	H	0.619	23
24		S	2	2	Q	49/50	1	1	4096	H	0.665	24
25		S	2	2	Q	49/50	2	1	4096	H	0.660	25
26		S	2	2	Q	9/10	0	1	16384	H	0.695	26
27		S	2	2	Q	9/10	1	1	16384	H	0.701	27
28		S	2	2	Q	9/10	2	1	16384	H	0.745	28
29		S	4	4	Q	49/50	0	1	4096	H	5.539	29
30		S	4	4	Q	49/50	1	1	4096	H	5.559	30
31		S	4	4	Q	49/50	2	1	4096	H	5.619	31
32		S	4	4	Q	9/10	0	1	4096	H	5.582	32
		S	4	4	Q	9/10	1	1	4096	H	5.576	
		S	4	4	Q	9/10	2	1	4096	H	5.583	
		S	8	6	Q	49/50	0	1	4096	H	20.344	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3	S	8	6	Q	49/50	1	1	4096	H	20.383	3
4	S	8	6	Q	49/50	2	1	4096	H	21.123	4
5	S	8	6	Q	9/10	0	1	16384	H	21.243	5
6	S	8	6	Q	9/10	1	1	16384	H	21.027	6
7	S	8	6	Q	9/10	2	1	16384	H	21.882	7
8	S	2	2	N	49/50	0	1	4096	H	1.304	8
9	S	2	2	N	49/50	1	1	4096	H	1.333	9
10	S	2	2	N	49/50	2	1	4096	H	1.292	10
11	S	2	2	N	9/10	0	1	16384	H	1.431	11
12	S	2	2	N	9/10	1	1	16384	H	1.389	12
13	S	2	2	N	9/10	2	1	16384	H	1.409	13
14	S	4	4	N	49/50	0	1	4096	H	9.111	14
15	S	4	4	N	49/50	1	1	4096	H	9.106	15
16	S	4	4	N	49/50	2	1	4096	H	9.140	16
17	S	4	4	N	9/10	0	1	4096	H	9.238	17
18	S	4	4	N	9/10	1	1	4096	H	9.087	18
19	S	4	4	N	9/10	2	1	4096	H	9.073	19
20	S	8	6	N	49/50	0	1	4096	H	25.945	20
21	S	8	6	N	49/50	1	1	4096	H	26.944	21
22	S	8	6	N	49/50	2	1	4096	H	25.865	22
23	S	8	6	N	9/10	0	1	16384	H	26.714	23
24	S	8	6	N	9/10	1	1	16384	H	30.501	24
25	S	8	6	N	9/10	2	1	16384	H	26.771	25
26	S	2	2	F	49/50	0	1	4096	H	0.780	26
27	S	2	2	F	49/50	1	1	4096	H	0.785	27
28	S	2	2	F	49/50	2	1	4096	H	0.747	28
29	S	2	2	F	9/10	0	1	16384	H	0.807	29
30	S	2	2	F	9/10	1	1	16384	H	0.809	30
31	S	2	2	F	9/10	2	1	16384	H	0.853	31
32											32

P: primary cache; S: signed reference extension. H: target attained; E: declared budget exhausted. Repetitions 0–2 have identical training and inference seeds, not independent

training draws. All 132 processes and their immutable cumulative driver checkpoints are included. The full JSON retains every per-stage construction, cache, gate and inference clock.

3. ALL FINAL ECONOMIC INTERVALS

TABLE II: Every final actual-cost, gain and target interval

Cohort	d	T	Method	q	Cost	Gain	Target contrast
P	2	2	W	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	W	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	E	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	E	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	Q	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	Q	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	N	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	N	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	F	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	F	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	4	4	E	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	E	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	Q	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	Q	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	N	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	N	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	8	6	E	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	E	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	Q	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	Q	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	N	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	N	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
S	2	2	W	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.224, -0.048]
S	2	2	W	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	E	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	E	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]

Cohort	d	T	Method	q	Cost	Gain	Target contrast
S	2	2	Q	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	Q	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	N	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	N	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	F	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	F	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	4	4	E	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	E	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	Q	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	Q	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	N	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	N	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	8	6	E	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	E	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	Q	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	Q	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	N	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	N	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]

Intervals are rounded outward and correspond to repetition zero; exact identities across repetitions are separately checked. A negative target-contrast upper endpoint licenses the stated prospective return. Gain intervals use paired path endpoints rather than subtraction of marginal cost intervals. Every intermediate look remains in the service archive.

4. ALL DATEWISE ERROR ACCOUNTS

TABLE III: All datewise directed gaps and enclosure components

Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$
P	2	2	W	49/50	0	0	0	2.195	0.118	2.230

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 2	2	W	49/50	0	1	27	0.607	0.054	0.660	2
3		P 2	2	W	9/10	0	0	0	2.195	0.118	2.230	3
4		P 2	2	W	9/10	0	1	27	0.607	0.054	0.660	4
5		P 2	2	W	9/10	1	0	0	1.481	0.033	1.486	5
6		P 2	2	W	9/10	1	1	93	0.344	0.033	0.377	6
7		P 2	2	E	49/50	0	0	0	2.195	0.118	2.230	7
8		P 2	2	E	49/50	0	1	27	0.607	0.054	0.660	8
9		P 2	2	E	9/10	0	0	0	2.195	0.118	2.230	9
10		P 2	2	E	9/10	0	1	27	0.607	0.054	0.660	10
11		P 2	2	E	9/10	1	0	0	1.481	0.033	1.486	11
12		P 2	2	E	9/10	1	1	93	0.344	0.033	0.377	12
13		P 4	4	E	49/50	0	0	0	4.727	0.062	4.729	13
14		P 4	4	E	49/50	0	1	0	3.705	0.058	3.706	14
15		P 4	4	E	49/50	0	2	0	2.614	0.054	2.615	15
16		P 4	4	E	49/50	0	3	15	0.638	0.054	0.691	16
17		P 4	4	E	49/50	1	0	0	4.530	0.041	4.530	17
18		P 4	4	E	49/50	1	1	0	3.535	0.038	3.536	18
19		P 4	4	E	49/50	1	2	0	2.474	0.035	2.474	19
20		P 4	4	E	49/50	1	3	90	0.636	0.043	0.679	20
21		P 4	4	E	9/10	0	0	0	4.727	0.062	4.729	21
22		P 4	4	E	9/10	0	1	0	3.705	0.058	3.706	22
23		P 4	4	E	9/10	0	2	0	2.614	0.054	2.615	23
24		P 4	4	E	9/10	0	3	15	0.638	0.054	0.691	24
25		P 4	4	E	9/10	1	0	0	4.530	0.041	4.530	25
26		P 4	4	E	9/10	1	1	0	3.535	0.038	3.536	26
27		P 4	4	E	9/10	1	2	0	2.474	0.035	2.474	27
28		P 4	4	E	9/10	1	3	90	0.636	0.043	0.679	28
29		P 8	6	E	49/50	0	0	0	7.047	0.021	7.047	29
30		P 8	6	E	49/50	0	1	0	6.079	0.023	6.079	30
31		P 8	6	E	49/50	0	2	0	5.047	0.024	5.047	31
32		P 8	6	E	49/50	0	3	0	3.946	0.027	3.946	32
		P 8	6	E	49/50	0	4	0	2.771	0.029	2.772	
		P 8	6	E	49/50	0	5	0	0.640	0.074	0.713	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 8 6		E 49/50	1	0		0	6.836	0.058	6.836	2
3		P 8 6		E 49/50	1	1		0	5.901	0.056	5.902	3
4		P 8 6		E 49/50	1	2		0	4.904	0.055	4.904	4
5		P 8 6		E 49/50	1	3		0	3.840	0.053	3.841	5
6		P 8 6		E 49/50	1	4		0	2.705	0.052	2.706	6
7		P 8 6		E 49/50	1	5		0	0.639	0.064	0.702	7
8		P 8 6		E 9/10	0	0		0	7.047	0.021	7.047	8
9		P 8 6		E 9/10	0	1		0	6.079	0.023	6.079	9
10		P 8 6		E 9/10	0	2		0	5.047	0.024	5.047	10
11		P 8 6		E 9/10	0	3		0	3.946	0.027	3.946	11
12		P 8 6		E 9/10	0	4		0	2.771	0.029	2.772	12
13		P 8 6		E 9/10	0	5		0	0.640	0.074	0.713	13
14		P 8 6		E 9/10	1	0		0	6.836	0.058	6.836	14
15		P 8 6		E 9/10	1	1		0	5.901	0.056	5.902	15
16		P 8 6		E 9/10	1	2		0	4.904	0.055	4.904	16
17		P 8 6		E 9/10	1	3		0	3.840	0.053	3.841	17
18		P 8 6		E 9/10	1	4		0	2.705	0.052	2.706	18
19		P 8 6		E 9/10	1	5		0	0.639	0.064	0.702	19
20		P 2 2		Q 49/50	0	0		0	2.195	0.118	2.230	20
21		P 2 2		Q 49/50	0	1		27	0.607	0.054	0.660	21
22		P 2 2		Q 9/10	0	0		0	2.195	0.118	2.230	22
23		P 2 2		Q 9/10	0	1		27	0.607	0.054	0.660	23
24		P 2 2		Q 9/10	1	0		0	1.361	0.033	1.388	24
25		P 2 2		Q 9/10	1	1		93	0.344	0.033	0.377	25
26		P 4 4		Q 49/50	0	0		0	4.727	0.062	4.729	26
27		P 4 4		Q 49/50	0	1		0	3.705	0.058	3.706	27
28		P 4 4		Q 49/50	0	2		0	2.614	0.054	2.615	28
29		P 4 4		Q 49/50	0	3		15	0.638	0.054	0.691	29
30		P 4 4		Q 49/50	1	0		0	4.530	0.041	4.530	30
31		P 4 4		Q 49/50	1	1		0	3.535	0.038	3.536	31
32		P 4 4		Q 49/50	1	2		0	2.474	0.035	2.474	32
		P 4 4		Q 49/50	1	3		90	0.636	0.043	0.679	
		P 4 4		Q 9/10	0	0		0	4.727	0.062	4.729	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 4	4	Q	9/10	0	1	0	3.705	0.058	3.706	2
3		P 4	4	Q	9/10	0	2	0	2.614	0.054	2.615	3
4		P 4	4	Q	9/10	0	3	15	0.638	0.054	0.691	4
5		P 4	4	Q	9/10	1	0	0	4.530	0.041	4.530	5
6		P 4	4	Q	9/10	1	1	0	3.535	0.038	3.536	6
7		P 4	4	Q	9/10	1	2	0	2.474	0.035	2.474	7
8		P 4	4	Q	9/10	1	3	90	0.636	0.043	0.679	8
9		P 8	6	Q	49/50	0	0	0	7.047	0.021	7.047	9
10		P 8	6	Q	49/50	0	1	0	6.079	0.023	6.079	10
11		P 8	6	Q	49/50	0	2	0	5.047	0.024	5.047	11
12		P 8	6	Q	49/50	0	3	0	3.946	0.027	3.946	12
13		P 8	6	Q	49/50	0	4	0	2.771	0.029	2.772	13
14		P 8	6	Q	49/50	0	5	0	0.640	0.074	0.713	14
15		P 8	6	Q	49/50	1	0	0	6.836	0.058	6.836	15
16		P 8	6	Q	49/50	1	1	0	5.901	0.056	5.902	16
17		P 8	6	Q	49/50	1	2	0	4.904	0.055	4.904	17
18		P 8	6	Q	49/50	1	3	0	3.840	0.053	3.841	18
19		P 8	6	Q	49/50	1	4	0	2.705	0.052	2.706	19
20		P 8	6	Q	49/50	1	5	0	0.639	0.064	0.702	20
21		P 8	6	Q	9/10	0	0	0	7.047	0.021	7.047	21
22		P 8	6	Q	9/10	0	1	0	6.079	0.023	6.079	22
23		P 8	6	Q	9/10	0	2	0	5.047	0.024	5.047	23
24		P 8	6	Q	9/10	0	3	0	3.946	0.027	3.946	24
25		P 8	6	Q	9/10	0	4	0	2.771	0.029	2.772	25
26		P 8	6	Q	9/10	0	5	0	0.640	0.074	0.713	26
27		P 8	6	Q	9/10	1	0	0	6.836	0.058	6.836	27
28		P 8	6	Q	9/10	1	1	0	5.901	0.056	5.902	28
29		P 8	6	Q	9/10	1	2	0	4.904	0.055	4.904	29
30		P 8	6	Q	9/10	1	3	0	3.840	0.053	3.841	30
31		P 8	6	Q	9/10	1	4	0	2.705	0.052	2.706	31
32		P 8	6	Q	9/10	1	5	0	0.639	0.064	0.702	32
		P 2	2	N	49/50	0	0	0	2.195	0.118	2.230	
		P 2	2	N	49/50	0	1	27	0.607	0.054	0.660	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2	P	2	2	N	9/10	0	0	0	2.195	0.118	2.230	2
3	P	2	2	N	9/10	0	1	27	0.607	0.054	0.660	3
4	P	2	2	N	9/10	1	0	0	1.481	0.033	1.486	4
5	P	2	2	N	9/10	1	1	93	0.344	0.033	0.377	5
6	P	4	4	N	49/50	0	0	0	4.727	0.062	4.729	6
7	P	4	4	N	49/50	0	1	0	3.705	0.058	3.706	7
8	P	4	4	N	49/50	0	2	0	2.614	0.054	2.615	8
9	P	4	4	N	49/50	0	3	15	0.638	0.054	0.691	9
10	P	4	4	N	49/50	1	0	0	4.530	0.041	4.530	10
11	P	4	4	N	49/50	1	1	0	3.535	0.038	3.536	11
12	P	4	4	N	49/50	1	2	0	2.474	0.035	2.474	12
13	P	4	4	N	49/50	1	3	90	0.636	0.043	0.679	13
14	P	4	4	N	9/10	0	0	0	4.727	0.062	4.729	14
15	P	4	4	N	9/10	0	1	0	3.705	0.058	3.706	15
16	P	4	4	N	9/10	0	2	0	2.614	0.054	2.615	16
17	P	4	4	N	9/10	0	3	15	0.638	0.054	0.691	17
18	P	4	4	N	9/10	1	0	0	4.530	0.041	4.530	18
19	P	4	4	N	9/10	1	1	0	3.535	0.038	3.536	19
20	P	4	4	N	9/10	1	2	0	2.474	0.035	2.474	20
21	P	4	4	N	9/10	1	3	90	0.636	0.043	0.679	21
22	P	8	6	N	49/50	0	0	0	7.047	0.021	7.047	22
23	P	8	6	N	49/50	0	1	0	6.079	0.023	6.079	23
24	P	8	6	N	49/50	0	2	0	5.047	0.024	5.047	24
25	P	8	6	N	49/50	0	3	0	3.946	0.027	3.946	25
26	P	8	6	N	49/50	0	4	0	2.771	0.029	2.772	26
27	P	8	6	N	49/50	0	5	0	0.640	0.074	0.713	27
28	P	8	6	N	49/50	1	0	0	6.836	0.058	6.836	28
29	P	8	6	N	49/50	1	1	0	5.901	0.056	5.902	29
30	P	8	6	N	49/50	1	2	0	4.904	0.055	4.904	30
31	P	8	6	N	49/50	1	3	0	3.840	0.053	3.841	31
32	P	8	6	N	49/50	1	4	0	2.705	0.052	2.706	32
	P	8	6	N	49/50	1	5	0	0.639	0.064	0.702	
	P	8	6	N	9/10	0	0	0	7.047	0.021	7.047	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$		
1												1	
2		P	8	6	N	9/10	0	1	0	6.079	0.023	6.079	2
3		P	8	6	N	9/10	0	2	0	5.047	0.024	5.047	3
4		P	8	6	N	9/10	0	3	0	3.946	0.027	3.946	4
5		P	8	6	N	9/10	0	4	0	2.771	0.029	2.772	5
6		P	8	6	N	9/10	0	5	0	0.640	0.074	0.713	6
7		P	8	6	N	9/10	1	0	0	6.836	0.058	6.836	7
8		P	8	6	N	9/10	1	1	0	5.901	0.056	5.902	8
9		P	8	6	N	9/10	1	2	0	4.904	0.055	4.904	9
10		P	8	6	N	9/10	1	3	0	3.840	0.053	3.841	10
11		P	8	6	N	9/10	1	4	0	2.705	0.052	2.706	11
12		P	8	6	N	9/10	1	5	0	0.639	0.064	0.702	12
13		P	2	2	F	49/50	0	0	0	2.195	0.118	2.230	13
14		P	2	2	F	49/50	0	1	27	0.607	0.054	0.660	14
15		P	2	2	F	9/10	0	0	0	2.195	0.118	2.230	15
16		P	2	2	F	9/10	0	1	27	0.607	0.054	0.660	16
17		P	2	2	F	9/10	1	0	0	1.481	0.033	1.486	17
18		P	2	2	F	9/10	1	1	93	0.344	0.033	0.377	18
19		S	2	2	W	49/50	0	0	30	0.319	0.086	0.405	19
20		S	2	2	W	49/50	0	1	27	0.607	0.054	0.660	20
21		S	2	2	W	9/10	0	0	30	0.319	0.086	0.405	21
22		S	2	2	W	9/10	0	1	27	0.607	0.054	0.660	22
23		S	2	2	E	49/50	0	0	30	0.319	0.086	0.405	23
24		S	2	2	E	49/50	0	1	27	0.607	0.054	0.660	24
25		S	2	2	E	9/10	0	0	30	0.319	0.086	0.405	25
26		S	2	2	E	9/10	0	1	27	0.607	0.054	0.660	26
27		S	4	4	E	49/50	0	0	31	0.503	0.246	0.749	27
28		S	4	4	E	49/50	0	1	31	0.464	0.227	0.691	28
29		S	4	4	E	49/50	0	2	28	0.400	0.197	0.596	29
30		S	4	4	E	49/50	0	3	15	0.638	0.054	0.691	30
31		S	4	4	E	9/10	0	0	31	0.503	0.246	0.749	31
32		S	4	4	E	9/10	0	1	31	0.464	0.227	0.691	32
		S	4	4	E	9/10	0	2	28	0.400	0.197	0.596	
		S	4	4	E	9/10	0	3	15	0.638	0.054	0.691	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2												2
3		S 8 6		E	49/50	0	0	26	0.604	0.418	1.021	3
4		S 8 6		E	49/50	0	1	24	0.588	0.407	0.994	4
5		S 8 6		E	49/50	0	2	24	0.562	0.388	0.950	5
6		S 8 6		E	49/50	0	3	20	0.521	0.358	0.879	6
7		S 8 6		E	49/50	0	4	14	0.453	0.310	0.762	7
8		S 8 6		E	49/50	0	5	0	0.640	0.074	0.713	8
9		S 8 6		E	9/10	0	0	26	0.604	0.418	1.021	9
10		S 8 6		E	9/10	0	1	24	0.588	0.407	0.994	10
11		S 8 6		E	9/10	0	2	24	0.562	0.388	0.950	11
12		S 8 6		E	9/10	0	3	20	0.521	0.358	0.879	12
13		S 8 6		E	9/10	0	4	14	0.453	0.310	0.762	13
14		S 8 6		E	9/10	0	5	0	0.640	0.074	0.713	14
15		S 2 2		Q	49/50	0	0	30	0.319	0.086	0.405	15
16		S 2 2		Q	49/50	0	1	27	0.607	0.054	0.660	16
17		S 2 2		Q	9/10	0	0	30	0.319	0.086	0.405	17
18		S 2 2		Q	9/10	0	1	27	0.607	0.054	0.660	18
19		S 4 4		Q	49/50	0	0	31	0.503	0.246	0.749	19
20		S 4 4		Q	49/50	0	1	31	0.464	0.227	0.691	20
21		S 4 4		Q	49/50	0	2	28	0.400	0.197	0.596	21
22		S 4 4		Q	49/50	0	3	15	0.638	0.054	0.691	22
23		S 4 4		Q	9/10	0	0	31	0.503	0.246	0.749	23
24		S 4 4		Q	9/10	0	1	31	0.464	0.227	0.691	24
25		S 4 4		Q	9/10	0	2	28	0.400	0.197	0.596	25
26		S 4 4		Q	9/10	0	3	15	0.638	0.054	0.691	26
27		S 8 6		Q	49/50	0	0	26	0.604	0.418	1.021	27
28		S 8 6		Q	49/50	0	1	24	0.588	0.407	0.994	28
29		S 8 6		Q	49/50	0	2	24	0.562	0.388	0.950	29
30		S 8 6		Q	49/50	0	3	20	0.521	0.358	0.879	30
31		S 8 6		Q	49/50	0	4	14	0.453	0.310	0.762	31
32		S 8 6		Q	49/50	0	5	0	0.640	0.074	0.713	32
		S 8 6		Q	9/10	0	0	26	0.604	0.418	1.021	
		S 8 6		Q	9/10	0	1	24	0.588	0.407	0.994	
		S 8 6		Q	9/10	0	2	24	0.562	0.388	0.950	

Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$
S 8 6	Q	9/10	0	3	20	0.521	0.358	0.879		
S 8 6	Q	9/10	0	4	14	0.453	0.310	0.762		
S 8 6	Q	9/10	0	5	0	0.640	0.074	0.713		
S 2 2	N	49/50	0	0	30	0.319	0.086	0.405		
S 2 2	N	49/50	0	1	27	0.607	0.054	0.660		
S 2 2	N	9/10	0	0	30	0.319	0.086	0.405		
S 2 2	N	9/10	0	1	27	0.607	0.054	0.660		
S 4 4	N	49/50	0	0	31	0.503	0.246	0.749		
S 4 4	N	49/50	0	1	31	0.464	0.227	0.691		
S 4 4	N	49/50	0	2	28	0.400	0.197	0.596		
S 4 4	N	49/50	0	3	15	0.638	0.054	0.691		
S 4 4	N	9/10	0	0	31	0.503	0.246	0.749		
S 4 4	N	9/10	0	1	31	0.464	0.227	0.691		
S 4 4	N	9/10	0	2	28	0.400	0.197	0.596		
S 4 4	N	9/10	0	3	15	0.638	0.054	0.691		
S 8 6	N	49/50	0	0	26	0.604	0.418	1.021		
S 8 6	N	49/50	0	1	24	0.588	0.407	0.994		
S 8 6	N	49/50	0	2	24	0.562	0.388	0.950		
S 8 6	N	49/50	0	3	20	0.521	0.358	0.879		
S 8 6	N	49/50	0	4	14	0.453	0.310	0.762		
S 8 6	N	49/50	0	5	0	0.640	0.074	0.713		
S 8 6	N	9/10	0	0	26	0.604	0.418	1.021		
S 8 6	N	9/10	0	1	24	0.588	0.407	0.994		
S 8 6	N	9/10	0	2	24	0.562	0.388	0.950		
S 8 6	N	9/10	0	3	20	0.521	0.358	0.879		
S 8 6	N	9/10	0	4	14	0.453	0.310	0.762		
S 8 6	N	9/10	0	5	0	0.640	0.074	0.713		
S 2 2	F	49/50	0	0	30	0.319	0.086	0.405		
S 2 2	F	49/50	0	1	27	0.607	0.054	0.660		
S 2 2	F	9/10	0	0	30	0.319	0.086	0.405		
S 2 2	F	9/10	0	1	27	0.607	0.054	0.660		

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All stages of repetition zero are retained. A maximum of sums need not equal the sum of maxima. Quantities are upper bounds rounded upward, not estimates of disjoint causal errors. Complete cell arrays contain the exact endpoints used by the gate.

5. READING AND PRESERVATION MAP

The active article is organized around construction and training, acquired-state improvement, prospective stopping and actual economic cost. The preceding main article is retained as an unchanged historical source and a rebuilt development document. The complete development article and complete proof supplement retain the full prior body and add the current theoretical and numerical sections. Historical labels and immutable section-source hashes are checked rather than relying on a statement that no results were removed.

The original resource-allocation, original-owner compiler, observation-price, recursive-preference, diffusion, temporal-self and game results remain available in those companions. The expected-cost sampling and linear action-kernel results in the active article do not claim that a nonlinear recursive utility or game has been experimentally validated by an investment-policy interval. A reading map is a distinction among assumptions and evidence, not abandonment of the original NBO research object.

REFERENCES